

PANEL 3 • OVERVIEW & MAP

PROJECT OVERVIEW

BACKGROUND CONTEXT

A university campus requires continuous spatial planning to optimize space and plan structural growth. Physical 3D models provide civil engineers and campus administrators with an intuitive spatial understanding of current assets.

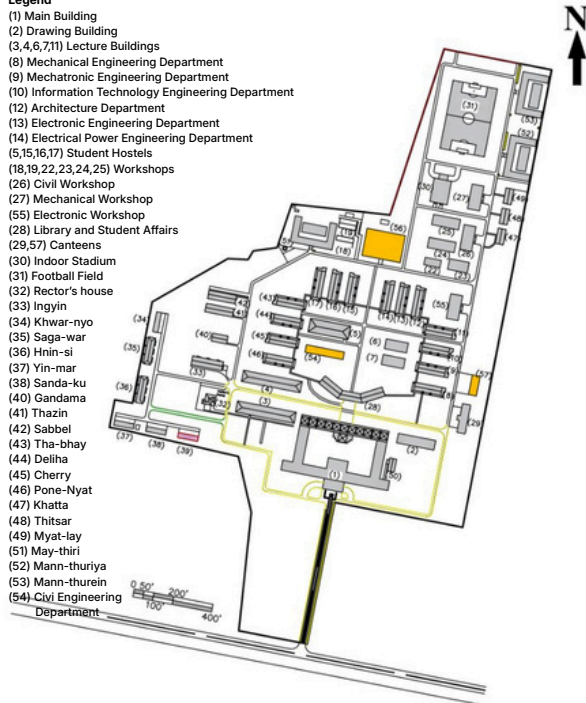
PROJECT SUMMARY

Detailed **1:120 scaled model** and 2D CAD representation of Technological University (Mandalay) across the **65.37-acre campus**.

MASTER CAMPUS BLUEPRINT MAP

Legend

- (1) Main Building
- (2) Drawing Building
- (3, 4, 6, 7, 11) Lecture Buildings
- (8) Mechanical Engineering Department
- (9) Mechatronic Engineering Department
- (10) Information Technology Engineering Department
- (12) Architecture Department
- (13) Electronic Engineering Department
- (14) Electrical Power Engineering Department
- (5, 15, 16, 17) Student Hostels
- (18, 19, 22, 23, 24, 25) Workshops
- (26) Civil Workshop
- (27) Mechanical Workshop
- (55) Electronic Workshop
- (28) Library and Student Affairs
- (29, 57) Canteens
- (30) Indoor Stadium
- (31) Football Field
- (32) Rector's house
- (33) Ingyin
- (34) Khwar-nyo
- (35) Saga-war
- (36) Hnin-si
- (37) Yin-mar
- (38) Sanda-ku
- (40) Gandama
- (41) Thazin
- (42) Sabbel
- (43) Tha-bhay
- (44) Deliha
- (45) Cherry
- (46) Pone-Nyat
- (47) Khatta
- (48) Thitsar
- (49) Myat-lay
- (51) May-thiri
- (52) Mann-thuriya
- (53) Mann-thurein
- (54) Civil Engineering Department



PANEL 2 • BACK COVER

PROJECT OBJECTIVES & VALUE

CORE OBJECTIVES

- ☐ **Precise Measurement:** Record accurate physical dimensions (length, breadth, height) across campus buildings using laser distance meters and manual tapes.
- ☐ **CAD Documentation:** Develop standardized 2D plans, elevations, and layout maps in AutoCAD and SketchUp.
- ☐ **Spatial Layout:** Position scaled 3D physical models to match actual geographic layout coordinates.
- ☐ **Institutional Legacy:** Create an enduring reference model of TUM campus infrastructure for future generations.

TEAMWORK & PEDAGOGICAL SKILLS |

Enhanced practical competencies in surveying, CAD drafting, architectural scaling, and physical model fabrication through collaborative engineering execution.



TECHNOLOGICAL UNIVERSITY (MANDALAY)

Aung Chan Thar Quarter, Pathein Gyi Township
Mandalay Region, Myanmar

DEPARTMENT OF CIVIL ENGINEERING

TECHNOLOGICAL UNIVERSITY (MANDALAY)

MODELLING OF TUM (Campus Infrastructure)

A Civil Engineering Capstone Project capturing the architectural footprint and spatial layout of TUM for academic reference and posterity.



KEY PROJECT SPECS

Campus Area: 65.37 Acres

Model Scale: 1:120

Primary Materials: PVC Sheets & Glass



Location:
Pathein Gyi Township, Mandalay

2026

PANEL 4 • METHODOLOGY

1. SURVEYING & CAD

DATA COLLECTION & FIELD SURVEY

Source Blueprint Data:

Master plan blueprints, building elevations, and structural layouts were sourced from the Regional Department of TUM.

Fieldwork & Verification:

Missing building elevations and physical dimensions were surveyed directly using:

- Laser Distance Meters:** High-altitude / depth precision.
- Measuring Tapes:** Ground-level perimeter verification.

2D CAD MODELLING

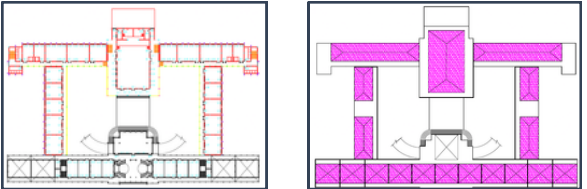
Software Tool: AutoCAD

Drafting Deliverables:

Detailed line plans, floorplans, structural sections, and exterior elevation profiles.

Purpose:

Serves both as the blueprint for physical cutting and as a digital archive for campus management.



Second and Roof Floor Plan of Main Building

PANEL 5 • CONSTRUCTION

2. FABRICATION & ASSEMBLY

MATERIALS & SPECS

Parameter	Specification / Detail
Selected Scale	1:120 (Optimal Detail Density)
Primary Shell	3mm & 5mm Polyvinyl Chloride (PVC) Sheets
Glazing	Clear Glass Plates for windows & doors
Finishing	ATM Spray Acrylic Lacquer (Matched to real buildings)

FABRICATION WORKFLOW

- Pattern Transfer:** Line drawings transferred directly onto PVC sheets.
- Precision Cutting:** Structural elements cut using precision utility tools.
- Detailing & Drilling:** Cordless drills for window/door apertures; glass inserted.
- Surface Finishing:** Spray coating applied to mirror actual structural colors.
- Final Assembly:** Structures anchored onto display base using real coordinates.

PANEL 6 • OUTCOMES

RESULTS & IMPACT

KEY FINDINGS

57 **Numbered Campus Assets**

53 individual buildings, indoor stadium, canteens & football field.

High Spatial Accuracy:

High-fidelity spatial alignment mirroring the official campus master plan.

Architectural Realism:

Realistic exterior texture, color matching, and structural proportions.

"By capturing the physical layout of TUM in both digital CAD(2D) format and a physical 1:120 scale model, this project provides a vital tool for spatial planning, future campus expansion, and administrative decision-making."

PLANNING TAKEAWAYS

- ✓ Aids administration in evaluating future expansion sites.
- ✓ Provides a physical visualization tool for visiting dignitaries and students.
- ✓ Demonstrates civil engineering survey techniques, CAD design, and architectural modeling integration.